

CS105 Practical 2 2024-2025

It is likely you may not need help with this practical exercise. However, if you have questions or have run in to difficulties, please drop in to the week 2 lab – Wed 1-3 in LT1201 & 1221.

The aim of this practical is to become familiar with creating objects and invoking their methods. Before starting this make sure you have first worked through the short Practical 1 exercise.

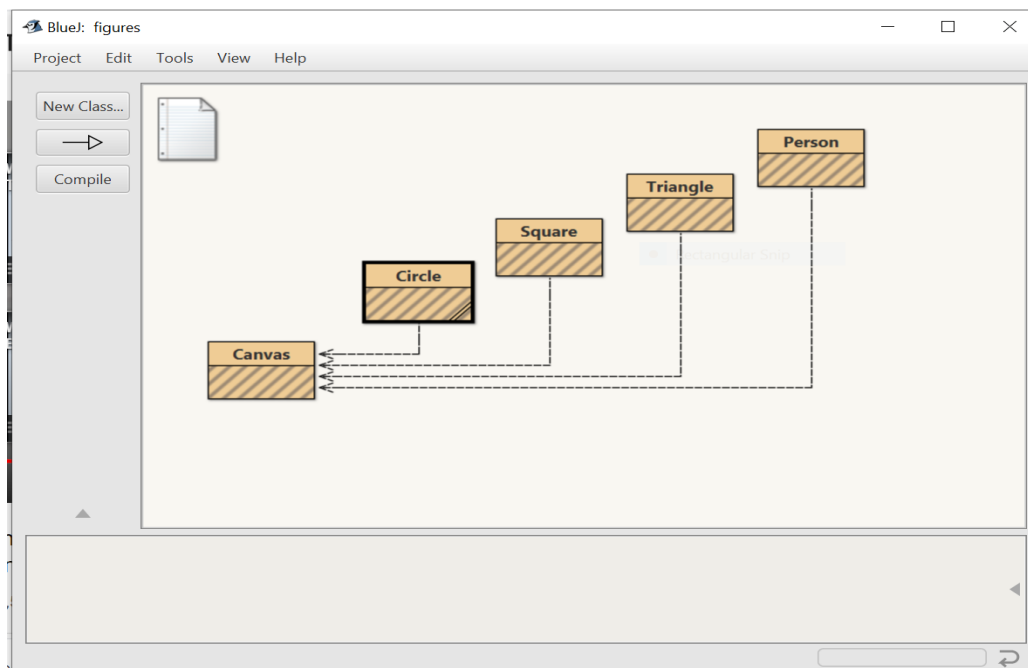
This practical exercise is not assessed. Write down your answers to the numbered questions so you may check them against sample solutions – available from MyPlace once you indicate that you have completed the practical on MyPlace.

Objects and Classes

Find (e.g. type BlueJ in the search box of Windows) and run the BlueJ App.

From the Project menu choose Open Project. Find and select the **figures** project from the Chapter 1 projects you saved in Practical 1.

If you have been successful you should see a screen that looks something like this:



If classes have diagonal slashes through them (as shown above) then you will need to Compile them – you can right-click then choose Compile for each class.

Now right click on the Circle class and choose **new Circle()** from the popup menu. Accept the default name in the 'name of instance' dialogue box. Click OK. You should see a red rectangle (with curved corners – we'll call it a roundtangle) towards the bottom of the screen (the object bench) labelled '**circle1**'. This represents a Circle object.

Create another Circle. Then create a square.

Calling methods

Right click on one of the circle objects (not the class). Choose **makeVisible** from the menu. You will notice other methods in the popup menu. Try invoking **moveRight** and **moveDown** a few times to move the circle about the screen.

1. What happens if you call **moveDown** twice?
2. What happens if you call **makeInvisible** twice?

Make the circle visible and make the other two objects visible too.

Parameters

Invoke the **moveHorizontal** method on one of your circles. You will see a dialogue box appear that prompts for input. Type in 60 and click OK. Try invoking the **moveVertical**, **slowMoveVertical** and **changeSize** methods.

3. How can you use **moveHorizontal** to move a circle 80 pixels to the left? (A pixel is a single dot on the screen).

Data Types

Invoke the **changeColor** method on one of your circle objects and enter the String "red" (**including quotes**). This should change the colour of the circle. (Note the American spelling of colour in the method name here – when a method is used, the spelling must match what has been set up). Now try changing the colour to be green.

4. Invoke the **changeColor** method and write the colour into the parameter field without the quotes. What happens?
5. This is a very simple example, and not many colours are supported. What happens when you specify a colour that is not known?

Multiple Instances

Create several circle objects on the object bench. You can do so by selecting **new Circle()** from the popup menu of the Circle class. Make them visible, then move them around on the screen by invoking appropriate methods. Make one big and yellow, make another one small and green. Try the other shapes too: create a few triangles, squares and person objects. Change their positions, sizes, and colours.

State

In BlueJ the state of an object can be inspected by selecting the **Inspect** option from the object's popup menu. Make sure you have several objects on the object bench and then inspect some of them in turn. Try changing the state of an object (for example by calling the **moveLeft** method) while the object inspector is open. You should see the values in the object inspector change – reopen the inspector if it has closed.

Another example: lab-classes

Close the **figures** project and open up the **lab-classes** project.

Create an object of class Student. You will notice that this time you are prompted not only for a name to refer to the instance, but also for some parameters. Fill them in before clicking on OK.

(Remember that parameters of type String must be written in double quotes.)

Return values

6. Create some Student objects. Call the **getName** method on each object. Explain what happens when you call getName.

Objects as parameters

Create an object of class LabClass. As the signature indicates, you need to specify the maximum number of students in that lab class (an integer). Enter your own choice of size, e.g. 30.

7. Call the **numberOfStudents** method of your lab class object. What does it do?

Look at the signature for the **enrollStudent** method. You will notice that the type of the expected parameter is Student. Make sure you have two or three students and a LabClass object on the object bench, then call the **enrollStudent** method of the LabClass object. With the input cursor in the dialogue entry field, click on one of the student objects on the object bench. This enters the name of the student object into the parameter field of the **enrollStudent** method. Click OK, and you have added the student to the LabClass object. Add one or more other students as well.

8. Call the `numberOfStudents` method again. Has the result changed from earlier?

Call the `printList` method of the `LabClass` object. You will see a list of all the students in that class printed to the BlueJ terminal window.

9. Open an inspector on one of your student objects. In addition to fields for name and id, there is a further field. What is it called?
10. Explore the protocol for student objects to find a method which updates the value in the above field and then call this method a couple of times on each of your student objects. What was the name of the method you used and what was the type of its parameter(s)?
11. Print a list of students in your lab class on to the terminal window. Which method did you use?
12. Use the inspector on your `LabClass` object to discover what fields it has. List the field names.

Set the instructor, room and time for the lab class using appropriate methods. Print the list for the lab class to the terminal window to check that these new details appear along with the student details.

When you have completed the above tasks you have finished the practical exercise. Please tick the box on MyPlace to indicate that you have done this. Sample solutions to the questions above should then become available for you to view on MyPlace so that you can check against your own answers.

Challenge Exercise

We haven't yet covered writing any Java code or editing and compiling in BlueJ but if you are feeling confident please try the following. Edit a bit of Java source code in the House project to include a moving sunset. Experiment and try. You can edit the source code by double clicking on the appropriate class icon. You can compile a class by right-clicking on the appropriate class icon and picking Compile from the menu. (Note: there is a BlueJ tutorial available from the Help menu.)

When you have finished

Close any projects that are open and quit BlueJ

Marking practical 2 as complete or getting help

If you have successfully completed this Practical (excluding the Challenge Exercise), please mark as completed on the CS105 MyPlace page. The practical is not assessed.

If you have questions or have run in to difficulties, please go to the drop-in week 2 lab (Wed 1-3 in LT1201 & 1221) to speak with one of the class demonstrators or ask on mattermost.

*Martin Goodfellow
Sept 2024*